A Periodic Table of Secret Handshakes: Enumerating and Profiling Recognition-Prefix Strategies for the Iterated Prisoner's Dilemma

Red-Cardigan

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Abstract

A recognition prefix, or secret handshake, lets a strategy in the iterated prisoner's dilemma identify a partner before committing to cooperation: it plays a fixed word, then cooperates only if the opponent has echoed that word. The handshake idea and the structural invariants of binary words are each classical, yet the prefix space itself has not been enumerated and profiled as an object. We enumerate the deterministic recognition-prefix strategies and apply a joint structural and evolutionary battery: minimal-automaton size, aperiodic autocorrelation, self-overlap, sequence complexity, and Moran-process fixation under three post-recognition regimes. Profiling reproduces the complete Barker spectrum, with peak-sidelobe-one prefixes occurring at exactly $k = 2, 3, 4, 5, 7, 11, 13$ and at no other length up to 13, a spectrum nowhere encoded in the tool. The minimal Moore machine of a length-k prefix takes every value in the band $[k + 2, 2k + 1]$ for all $k \leq 11$. A single cooperative bit decides whether a handshake can invade a population of defectors: the all-defect prefix sits below neutral drift, while one C bit raises fixation above neutral. Mimic-resistance is structure-independent under cooperate-forever recognition, where a forger always invades, but vigilant grim and tit-for-tat regimes reverse this for every handshake across all 36 tested parameter cells. The enumeration tool and all data are released open source for reproduction and extension.

1 Introduction

Cooperation in the iterated prisoner's dilemma (IPD) is fragile: a population of cooperators is open to invasion by defectors who take the tempting payoff and give nothing back (Axelrod, 1984). One escape is recognition. If a strategy can identify a like-minded partner before it commits to cooperation, it can extend cooperation to kin and withhold it from everyone else. The classical device for this is the secret handshake (Robson, 1990): a strategy plays a fixed sequence of moves, watches whether the opponent plays the same sequence, and cooperates only on a match. The handshake is a recognition prefix, a word $p \in \{C, D\}^k$ that gates the strategy's later behaviour.

Recognition prefixes are not a curiosity. They emerge spontaneously when finite- state strategies are evolved in the Moran process: handshake-like prefixes appear in the machines that survive, and they sharpen the line between strategies that invade defectors and strategies that do not (Knight et al., 2018). The prefix is also a natural unit of strategy complexity. A strategy realised by a finite automaton has a minimal-state form, and the number of states is the standard measure of how complicated the strategy is (Rubinstein, 1986; Abreu and Rubinstein, 1988); a handshake is among the simplest constructions that still implements conditional cooperation, and low-complexity structure is where direct reciprocity tends to settle (Press and Dyson, 2012; Hilbe et al., 2018).

The handshake concept and the mathematics it touches are both well established, and they have been developed apart from each other. The minimal automaton of a strategy and its computation by Myhill–Nerode equivalence and partition refinement are textbook (Nerode, 1958; Hopcroft, 1971; Rubinstein, 1986). The autocorrelation properties of binary words, and the lengths at which a word has a peak sidelobe of one, are the subject of the Barker-code literature (Barker, 1953). The self-overlap structure of words, including which words have no proper border, has a combinatorial theory of its own (Guibas and Odlyzko, 1981). What has not been done is to treat the recognition-prefix space as a single object: to enumerate it exhaustively and to profile each prefix with all of these invariants at once, together with how the gated strategy fares evolutionarily. We do that here.

We are precise about the novelty. The handshake mechanism is from Robson (1990); the emergence of recognition in evolved machines and the invasion-of-defectors test are from Knight et al. (2018); every structural invariant we compute is a known primitive. The contribution is the synthesis: the exhaustive enumeration of the prefix space and its profiling under a joint structural and evolutionary battery, which lets us state where structure lives in the space, how it scales, and when it matters for cooperation. We claim the catalogue and what it reveals, not the concept or the primitives.

- **An exhaustive enumeration with a joint invariant battery.** We enumerate all deterministic recognition-prefix strategies and compute, for each, a structural profile (minimal-automaton size, aperiodic autocorrelation, self-overlap and borderedness, and sequence complexity) and an evolutionary profile (Moran-process fixation under three post-recognition regimes). The enumeration is deterministic and the fixation probabilities are closed form, so the catalogue is exactly reproducible (section 3).
- **A reproducible family taxonomy.** We assign each prefix to one of six rule-based families by priority predicates over its invariants, and we cross-check the assignment against unsupervised clustering of the invariant vectors (table 1, section 5).
- **Findings.** Profiling recovers the complete Barker spectrum as a validation of the autocorrelation pipeline (table 2); the minimal automaton occupies an exact band $[k+2,\, 2k+1]$ for every enumerated length $k \leq 11$ (table 3); a single cooperative bit is the threshold for a handshake to invade a defector population (table 5); and mimic-resistance is structure-independent under cooperate-forever recognition but is restored for every handshake by the vigilant grim and tit-for-tat regimes (table 6, table 7).

2 Preliminaries

Stage game and payoffs. Each round both players choose an action from $\{C, D\}$, where C is cooperate and D is defect. The payoffs are $R = 3$ for mutual cooperation, $P = 1$ for mutual defection, $T = 5$ to a defector facing a cooperator, and $S = 0$ to the exploited cooperator. These satisfy $T > R > P > S$ and $2R > T + S$, the standard prisoner's-dilemma ordering (Axelrod, 1984). A match runs for n rounds ($n = 200$ by default), and a player's score is the mean payoff per round.

Recognition-prefix strategies. A handshake is a recognition prefix $p \in \{C, D\}^k$ of length $k \geq 1$. The gated strategy H_p with post-recognition regime ρ behaves as follows. For the first k rounds it plays p regardless of what the opponent does. After round k it compares the opponent's first k actions with p. If they are equal, the opponent is recognised and H_p enters regime ρ; otherwise H_p defects on every remaining round. The first k rounds are the recognition phase, and the prefix doubles as the signal H_p emits and the password it requires.

Post-recognition regimes. We study three regimes $\rho \in \{\textsc{coop}, \textsc{grim}, \textsc{tft}\}$. Under $\textsc{coop}$ the strategy cooperates on every round after recognition; this is the classical secret handshake of Robson (1990), which trusts a recognised partner unconditionally. Under GRIM the strategy cooperates with a recognised partner until that partner defects once, then defects on every remaining round. Under TFT the strategy cooperates on the first post- recognition round and thereafter copies the partner's previous action. The regimes differ only in how a recognised partner is treated; recognition itself is identical.

The mimic. A mimic of p is the cheat that forges the signal: it plays p through the recognition phase, so H_p recognises it, and then defects on every remaining round. The mimic is the natural adversary for the handshake, and whether it can invade a population of handshakers measures how much the recognition gate is worth.

Minimal Moore machine. We realise H_p as a Moore machine, a finite-state automaton that emits an action in each state and transitions on the opponent's action. Two states are equivalent when they induce the same behaviour against every continuation, the Myhill–Nerode relation (Nerode, 1958); collapsing equivalent states gives the unique minimal machine, which we compute by Hopcroft partition refinement (Hopcroft, 1971). We write $m(p)$ for the number of states of this minimal machine. It is the Rubinstein measure of strategy complexity (Rubinstein, 1986), the size of the smallest automaton that plays H_p.

Aperiodic autocorrelation. Map p to its ± 1 image $a \in \{-1, +1\}^N$ with $N = k$ by sending C to $+1$ and D to -1. The aperiodic autocorrelation at shift v is $c_v = \sum_j a_j a_{j+v}$, summed over the overlapping positions. The sidelobes are the c_v for $v \neq 0$, the peak sidelobe is $\max_{v \neq 0} |c_v|$, and the merit factor is $N^2/(2 \sum_{v \neq 0} c_v^2)$. A word is a Barker code when its peak sidelobe is at most 1 (Barker, 1953), the strongest possible flatness, and such codes exist only at a short list of lengths (Turyn and Storer, 1961).

Self-overlap and bordered words. A border of p is a proper prefix that is also a suffix. A word is unbordered, or bifix-free, when it has no proper border, so it cannot align with a shifted copy of itself except at full overlap. Borders and the correlation polynomial that records them are the Guibas–Odlyzko theory of string overlaps (Guibas and Odlyzko, 1981); the number of unbordered binary words of each length is the sequence (OEIS Foundation Inc., 2024).

Sequence complexity. We summarise the incompressibility of p by its LZ76 production complexity, the number of distinct factors a left-to-right parse produces (Lempel and Ziv, 1976), and by its linear complexity, the length of the shortest linear-feedback shift register that generates p, computed by the Berlekamp–Massey algorithm (Massey, 1969). Together these separate structured prefixes from near-incompressible ones.

Moran process. We place H_p in a finite population of size N ($N = 20$ by default) evolving under the Moran process (Moran, 1958): each step one individual reproduces with probability proportional to fitness and one is removed at random. Fitness is mapped from match score with selection intensity w ($w = 0.1$ by default). The fixation probability of a single mutant in a resident population follows the closed-form constant-selection formula (Nowak et al., 2004), which we evaluate exactly with no simulation. The neutral-drift baseline, the fixation probability of a selectively neutral mutant, is $1/N$.

3 The atlas pipeline

The atlas is produced by a single deterministic pipeline that takes a length bound and returns, for every handshake within that bound, a vector of invariants drawn from three views of the recognition prefix. No step uses randomness, and every numeric column is reproducible to the last digit from the prefix alone.

3.1 Enumeration

For a fixed length k the prefix alphabet is $\{C, D\}$, so the stratum of length-k handshakes has 2^k members. We enumerate each stratum exhaustively in big-endian order, reading C as the low symbol and D as the high symbol within a fixed-width binary counter, and concatenate strata by increasing k. The order is total and canonical, so two runs of the pipeline visit handshakes in the same sequence and assign the same index to each. Structural invariants are computed for every handshake up to $k \leq 13$, which is $\sum_{k=1}^{13} 2^k = 16{,}382$ handshakes. The evolutionary view, which requires a pairwise match per handshake, is computed up to $k \leq 8$, which is $\sum_{k=1}^{8} 2^k = 510$ handshakes. Both bounds are exhaustive within their stratum: no sampling, no pruning.

3.2 The joint invariant battery

The invariants are grouped into three views, matching the three ways a recognition prefix can be read: as a string, as a machine, and as a player.

String view. The prefix p is mapped to its ± 1 image and profiled as a finite binary sequence. We record the aperiodic autocorrelation sidelobes $c_v = \sum_j a_j a_{j+v}$, the peak sidelobe $\max_v |c_v|$, and the merit factor $N^2/(2\sum_v c_v^2)$; a handshake is a Barker code when its peak sidelobe is at most 1 (Barker, 1953; Turyn and Storer, 1961). From the Guibas–Odlyzko self-overlap (correlation) polynomial we read whether p has a proper border, that is a proper prefix equal to a suffix; a prefix with no proper border is unbordered (bifix-free) (Guibas and Odlyzko, 1981). Compressibility is measured three ways: LZ76 parsing complexity (Lempel and Ziv, 1976), linear complexity via the Berlekamp–Massey recurrence (Massey, 1969), and factor complexity. These columns are exact integer or rational functions of p.

Automaton view. The gated strategy H_p is realised as a Moore machine. The construction has a spine of k states that play out p and check the opponent against it round by round, a post-recognition regime ρ, and a defect-forever sink, giving an unminimised machine of $2k+2$ states. We then minimise by partition refinement: states are split until no block contains two states with different outputs or with a transition leaving the block on some symbol, which is exactly the Myhill–Nerode quotient (Nerode, 1958; Hopcroft, 1971). The resulting state count $m(p)$ is the Rubinstein strategy-complexity measure of H_p (Rubinstein, 1986). Because the refinement is canonical, $m(p)$ is a deterministic function of p and ρ.

Evolutionary view. For the evolutionary columns each handshake is played in a deterministic pairwise match against a fixed reference set under the payoffs $R = 3$, $S = 0$, $T = 5$, $P = 1$. Since both players are deterministic, the round-by-round play and therefore the exact mean per-round payoff over n rounds ($n = 200$ by default) are fixed, with no Monte Carlo estimate. These mean payoffs are the entries of a constant-selection fitness comparison, which we feed to the closed-form fixation probability of a single mutant in a finite Moran population (Moran, 1958; Nowak et al., 2004), with default population $N = 20$ and selection intensity $w = 0.1$; the neutral-drift baseline is $1/N$. Two tests use this machinery. The *invade-AllD* test computes the fixation probability of H_p under the COOP regime invading a population of unconditional defectors, isolating the cost of carrying a recognition prefix. The *mimic-resistance* test computes the fixation probability of a mimic of p (which plays p then defects forever) against a resident population of H_p, run once per post-recognition regime $\rho \in \{\text{COOP}, \text{GRIM}, \text{TFT}\}$, to ask whether the regime punishes a forged signal.

Algorithm 1 The atlas pipeline for length bound K

for $k = 1$ to K **do**
  for each $p \in \{C, D\}^k$ in big-endian order **do**
    String: compute autocorrelation sidelobes, peak sidelobe, merit factor, Barker flag; self-overlap and unbordered flag; LZ76, linear, and factor complexity.
    Automaton: build the $2k+2$-state Moore machine for H_p; partition-refine to obtain $m(p)$.
    if $k \leq 8$ **then**
      Evolutionary: run the deterministic match for exact mean payoffs; evaluate the closed-form Moran fixation for invade-AllD and for mimic-resistance under each ρ.
    end if
    emit the invariant row for p.
  end for
end for

Table 1: The six handshake families, defined as priority predicates over the invariants. A handshake is assigned to the first family whose predicate it satisfies, top to bottom.

Family	Defining predicate	Characterisation
Constant	$p \in \{C^k, D^k\}$	one repeated symbol
Alternating	$p = (CD)^*$ or $(DC)^*$ truncated	period-two pattern
Barker	peak autocorrelation sidelobe ≤ 1	flat-sidelobe code
Unbordered	no proper border (bifix-free)	no prefix-suffix overlap
Self-similar	low LZ76 / linear complexity, not above	compressible, repetitive
Generic	none of the above	incompressible bulk

3.3 Determinism and reproducibility

Every stage is a deterministic function of the prefix and a small fixed parameter set. The enumeration order is canonical, the minimisation is canonical, the match play is deterministic, and the fixation probabilities are evaluated in closed form rather than simulated. Re-running the pipeline reproduces every number exactly; there is no random seed to fix.

3.4 Scope of the distance column

One natural invariant is deliberately absent at the per-handshake level. Within a full length-k stratum every prefix has a neighbour at Hamming distance 1 (flip any single bit), so the minimum distance from a handshake to the rest of its stratum is trivially 1 and carries no information. We therefore treat Hamming distance as a set-level quantity, useful for separating families or code sets, and not as a per-handshake column in the atlas.

4 A taxonomy of handshake families

The invariants induce a coarse classification of handshakes into six families. Each family is a predicate over the invariant vector, and a handshake is assigned to the first family whose predicate it satisfies, in the order Constant, Alternating, Barker, Unbordered, Self-similar, Generic. The first-match rule makes the assignment a total function, with Generic as the catch-all for prefixes that no earlier predicate selects. Table 1 states the predicates.

Table 2: Barker handshakes per length (peak autocorrelation sidelobe ≤ 1), counted over the enumerated space for $k \leq 13$. Lengths $6, 8, 9, 10, 12$ contribute none. The $k = 13$ set includes $CCCCCDDCCDCDC$.

Length k	Barker handshakes
2	4
3	4
4	8
5	4
7	4
11	4
13	4

4.1 The taxonomy recovers the geometry

The families are rule-based, so it is fair to ask whether they correspond to any structure in the invariants themselves or merely reflect the thresholds chosen. Clustering the standardised invariant vectors directly answers this.

Observation 1 (k-means recovers the rule families)**.** *For $k \leq 8$, k-means over the standardised joint-invariant vectors recovers the rule-based families as distinct clusters: one cluster is* 100% *Constant, one is* 85% *Barker, two are* 88–100% *Unbordered, and one is* 76% *Self-similar. The unsupervised geometry therefore aligns with the priority predicates, rather than the predicates imposing arbitrary cuts on an otherwise structureless space.*

4.2 A thin structured shell

The structured families do not grow with the space. There are always exactly two Constant handshakes, C^k and D^k, at every length, and for $k \geq 3$ there are always exactly two Alternating handshakes. The stratum total, by contrast, doubles at each length. The share of these structured families therefore tends to zero: the recognisable, compressible prefixes form a thin shell inside an exponentially growing bulk of Generic handshakes. The same picture holds for the unbordered fraction, which decreases along the length axis and converges from above toward the known bifix-free density ≈ 0.2678 (OEIS Foundation Inc., 2024); the special structure is a vanishing minority of the recognition space.

5 Results

5.1 Validation: the Barker spectrum

The autocorrelation module recovers the complete known Barker spectrum without being told what it is. Run blind over every handshake with $k \leq 13$, the module reports peak sidelobe ≤ 1 at exactly $k = 2, 3, 4, 5, 7, 11, 13$ and at no other length up to 13 (table 2). This matches the established spectrum of binary Barker sequences (Barker, 1953; Turyn and Storer, 1961). The spectrum is nowhere encoded in the software: the module computes aperiodic autocorrelation sidelobes from the ± 1 image of each prefix and applies the peak-≤ 1 test, so the agreement is an emergent property of the enumeration rather than a lookup.

Because a classical invariant from coding theory falls out of an enumeration built for an unrelated purpose, the autocorrelation pipeline is calibrated: subsequent structural readings from the same machinery can be taken at face value.

Table 3: Minimal-automaton size band. For each $k \leq 11$, $m(p)$ takes every value in $[k+2, 2k+1]$; the unminimised construction uses $2k+2$ states.

Length k	Minimum $m(p)$	Maximum $m(p)$	Unminimised
1	3	3	4
2	4	5	6
3	5	7	8
4	6	9	10
5	7	11	12
6	8	13	14
7	9	15	16
8	10	17	18
9	11	19	20
10	12	21	22
11	13	23	24

5.2 The minimal-automaton band

The minimal-automaton size of a gated strategy is confined to a narrow band, well below the naive construction. Building H_p directly costs $2k+2$ states. After minimisation by partition refinement (Myhill–Nerode), the realised count $m(p)$ never reaches that bound.

Observation 2 (Automaton band). *For every k with $1 \leq k \leq 11$, $m(p)$ ranges over exactly the interval $[\,k+2,\ 2k+1\,]$, attaining every integer value in it, against the unminimised construction of $2k+2$ states. The minimum $k+2$ is reached by D-tailed prefixes (e.g. DDDDDD, CDDDDD); the maximum $2k+1$ by C-heavy prefixes (e.g. CCCCCC, CCCDDC).*

table 3 lists the band endpoints. The C/D asymmetry between the extremes has a mechanical source: the post-recognition regime emits C on a recognised partner while the defect-forever terminal emits D, so a D-tailed prefix can merge its closing rounds into the defection terminal, collapsing states, whereas a C-heavy prefix cannot fold into either terminal and keeps its rounds distinct.

Conjecture 1 (Band extends). *The interval $[\,k+2,\ 2k+1\,]$ describes the full range of $m(p)$ for all $k \geq 1$ (verified by enumeration for $k \leq 11$).*

5.3 Self-overlap density

The fraction of unbordered handshakes falls with length and converges from above to the bifix-free density. A prefix is unbordered when no proper prefix equals a suffix (no proper border); table 4 gives the fraction at each length. The values decrease, are equal in consecutive pairs, and approach the known limiting density ≈ 0.2678 of bifix-free binary words (OEIS Foundation Inc., 2024).

The approach from above means unbordered prefixes, the ones that admit no self-overlap and so resist accidental partial matches, remain a constant-share minority rather than vanishing: at $k = 13$ roughly 27% of handshakes are still bifix-free.

5.4 Probe cost and the cooperative-bit threshold

A single cooperative bit decides whether a handshake can invade a population of defectors. We measure the fixation probability of H_p under the COOP regime invading an all-defector population, with neutral drift at $1/20 = 0.05$.

Table 4: Fraction of unbordered (bifix-free) handshakes by length, $k \leq 13$. The sequence decreases monotonically toward the limiting bifix-free density ≈ 0.2678.

Length k	Unbordered fraction
1	1.0000
2	0.5000
3	0.5000
4	0.3750
5	0.3750
6	0.3125
7	0.3125
8	0.2891
9	0.2891
10	0.2773
11	0.2773
12	0.2725
13	0.2725

Table 5: Mean fixation probability of H_p (COOP) invading AllD, by number of C bits in the prefix. Neutral drift is 0.05. The all-D prefix sits below neutral; one C bit lifts it above.

Length k	0 C-bits	≥ 1 C-bit
4	0.0095	0.0773
6	0.0096	0.0750–0.0770
8	0.0098	0.0768

The all-defect prefix is exploited rather than rewarded (table 5). At $k = 6$ it fixes at 0.0096, below neutral, because resident defectors reproduce the all-D prefix exactly: H_p misrecognises them as kin and then cooperates into exploitation. The mechanism shows in the per-pair payoffs: against AllD, $DDDD$ earns 0.020 while $CDDD$ earns 0.995. One C bit makes the prefix unforgeable by pure defectors and flips H_p to invasion-favoured at ≈ 0.077. The pattern is consistent at $k = 4$ ($0.0095 \to 0.0773$) and $k = 8$ ($0.0098 \to 0.0768$).

5.5 Mimic-resistance is structure-independent; vigilance fixes it

No prefix structure buys resistance to mimicry; the post-recognition regime does. A mimic of p plays the prefix, is recognised, then defects forever. We state the reading as a hypothesis that the enumeration could have refuted.

Observation 3 (Regime, not structure)**.** *Under* COOP, *a mimic invades for every handshake tested, fixing at $\approx 0.15 > 0.05$. Under* GRIM *and* TFT, *the mimic is resisted for every handshake tested, fixing at $\approx 0.015 < 0.05$. The direction does not vary with k or with any structural invariant of p (verified for $k \leq 8$).*

table 6 gives the regime contrast. The uniformity is the point: had even one family of prefixes (Barker, unbordered, or otherwise) resisted a mimic under COOP, the hypothesis would have failed. It does not. The COOP regime reproduces the tension identified by Robson (1990), where a recognition signal that triggers unconditional cooperation is forgeable and so invadable. The fix is behavioural rather than structural: a vigilant regime lets the mimic steal one round, then collapses the interaction to mutual defection (P each), so the forgery yields nothing sustained. This is consistent with the empirical finding of Knight et al. (2018) that self-recognition handshakes emerge in evolved machines and resist invasion when paired with conditional play.

Table 6: Mimic fixation probability by post-recognition regime, default $N = 20$, $w = 0.1$, $n = 200$. Neutral drift is 0.05. Direction is uniform across all handshakes for $k \leq 8$.

Regime	Mimic fixation	Outcome
COOP	≈ 0.15	mimic invades (> neutral)
GRIM	≈ 0.015	mimic resisted (< neutral)
TFT	≈ 0.015	mimic resisted (< neutral)

Table 7: Mimic fixation across representative cells of the 36-cell grid. The COOP/GRIM ordering holds in every cell; the margin grows with selection.

N	w	n	COOP	GRIM	neutral $1/N$
10	0.01	50	0.109	0.095	0.100
100	0.5	1000	0.339	0.000	0.010

6 Robustness

The regime result survives the full parameter grid. Sweeping $N \in \{10, 20, 50, 100\}$, $w \in \{0.01, 0.1, 0.5\}$, and $n \in \{50, 200, 1000\}$ gives 36 cells, and in all 36 the COOP mimic fixes above neutral while GRIM and TFT fix below it. The margin scales with selection strength rather than reversing. table 7 reports representative cells: at the weakest setting (N=10, w=0.01, n=50) the gap is small (COOP 0.109 vs. GRIM 0.095, neutral 0.100); at the strongest (N=100, w=0.5, n=1000) it is wide (COOP 0.339 vs. GRIM 0.000).

The structural conclusions are not cutoff artefacts. The automaton band (Observation 2) is confirmed for all $k \leq 11$ and the unbordered density (table 4) for all $k \leq 13$, in each case with the pattern stable across the final lengths enumerated rather than drifting near the boundary.

7 Related work

The secret handshake, a recognition signal that licenses cooperation among bearers, originates with Robson (1990); Knight et al. (2018) show such self-recognition mechanisms emerge in finite-state strategies evolved under the Moran process and survive an invasion-of-defectors test. Strategy complexity for repeated-game automata was set out by Rubinstein (1986) and Abreu and Rubinstein (1988), who tie equilibrium structure to the number of machine states; $m(p)$ is the same measure applied to gated strategies. Low-complexity strategy structure and its classification by payoff geometry appear in Press and Dyson (2012), on strategies that fix an opponent's payoff, and Hilbe et al. (2018), on the partners-versus-rivals split of memory-one play. The simulation ecosystem for the iterated prisoner's dilemma runs from the original tournaments of Axelrod (1984) to the reproducible open framework of Knight et al. (2016). The classical invariants we read off each prefix come from separate fields: Barker sequences and the peak-sidelobe bound (Barker, 1953; Turyn and Storer, 1961), string overlaps and bordering (Guibas and Odlyzko, 1981), and the complexity measures of Lempel and Ziv (1976) and Massey (1969). The contribution here is the synthesis: a single exhaustive enumeration that profiles each recognition prefix jointly by its coding-theoretic, combinatorial, automata-theoretic, and evolutionary invariants, and reads the cross-field structure off one table. No individual invariant is new.

8 Limitations

Construct validity. We operationalise mimic-resistance as the fixation probability of a prefix-forging mutant falling below neutral drift. This captures the evolutionary question directly, but it does not capture every sense of resistance: a mimic could persist as a polymorphism without fixing, or impose costs short of takeover. We report fixation because it is the standard finite-population summary (Nowak et al., 2004; Moran, 1958) and because the COOP/GRIM contrast is unambiguous under it.

External validity. The numerical results hold within the enumerated regime: structural invariants for $k \leq 13$, evolutionary invariants for $k \leq 8$. Behaviour at unbounded prefix length is conjectural (Conjecture 1). The evolutionary model is a well-mixed Moran process with fixed population N and the specific payoff matrix $R = 3, S = 0, T = 5, P = 1$; structured populations, alternative update rules, or a different payoff matrix may shift the margins, and we sweep N, w, and n but not these.

Evolutionary scope. Fixation probabilities are enumerated only for $k \leq 8$ (510 handshakes), so claims about the regime contrast are verified there and not beyond.

Proof status. The automaton band and the unbordered-density limit are computational observations over the enumerated set, not proofs. Conjecture 1 in particular is unproven.

9 Conclusion

The reading that organises the rest follows from the regime result: since no prefix structure resists a mimic, the recognition signal is irreparably forgeable. Any prefix can be replayed by a cheat that then defects, and no choice of k, no Barker prefix, no unbordered prefix changes this. Defence against forgery therefore cannot live in the handshake. It has to live in the post-recognition strategy, where a vigilant regime makes recognition cheap to grant and defection cheap to resume. The recognition prefix is best read as an addressing scheme, not a security mechanism.

Two open problems remain concrete. First, prove that $m(p)$ occupies exactly $[\,k+2,\ 2k+1\,]$ for all k, upgrading Conjecture 1 from an enumeration to a theorem about the minimised gated construction. Second, sweep alternative update rules and structured populations to test whether the COOP/GRIM ordering, uniform across 36 well-mixed cells, holds once the population is no longer well mixed.

Data and code availability. The enumeration tool, the handshake atlas, and the full reproduction scripts are open-source, written in Python with NumPy. The pipeline is deterministic and uses no random number generator: fixation probabilities are computed in closed form, so every figure and table is exactly reproducible. Code and atlas are available at `https://github.com/Red-Cardigan/handshake-atlas`; a tagged release with a DOI will accompany the submission.

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